

The Race between Man and Machine

Automation, new tasks, wages, and factor shares

Repository 6 — Artificial Intelligence and Economic Modeling

Acemoglu & Restrepo (AER, 2018)

Economic analysis: published AER 108(6), 1488–1542

Lean source: NBER Working Paper 22252, revised June 2017

Research question and core answer

Question

Does automation necessarily reduce wages and the labor share?

- Automation must first change *effective* task allocation: I is technological capability, while I^* is equilibrium automation.
- If automation reallocates tasks, the labor share falls through **displacement**.
- The wage is conditional because displacement competes with a positive **productivity effect**.
- New labor-intensive tasks create a **reinstatement effect**.

Bottom line

The labor-share effect is one-sided under the relevant static conditions; the wage effect is not.

Continuum of tasks and final output

Tasks are indexed on a continuum:

$$i \in [N - 1, N].$$

Each task produces an intermediate $y(i)$, and the final good aggregates all tasks with constant elasticity:

$$Y = \left(\int_{N-1}^N y(i)^{\frac{\sigma-1}{\sigma}} di \right)^{\frac{\sigma}{\sigma-1}}.$$

- A task is an activity required to produce final output.
- The interval has unit measure; N locates the frontier of task content.
- Task allocation is therefore an endogenous margin of aggregate production.

Task technologies and comparative advantage

For tasks that machines can perform ($i \leq I$), either factor may be used:

$$y(i) = [k(i) + \gamma(i)l(i)]^{1-\eta} q(i)^\eta.$$

For nonautomatable tasks ($i > I$), labor is required:

$$y(i) = [\gamma(i)l(i)]^{1-\eta} q(i)^\eta.$$

- $\gamma(i)$ is increasing: labor has comparative advantage in higher-index tasks.
- Effective factor costs are R for capital and $W/\gamma(i)$ for labor.
- An automatable task uses capital when $R < W/\gamma(i)$.

Technological frontier I versus effective frontier I^*

Define the cost frontier by

$$\frac{W}{R} = \gamma(\tilde{I}).$$

Equilibrium automation is

$$I^* = \min\{I, \tilde{I}\}.$$

Technology constrained

$I < \tilde{I}$, hence $I^* = I$. A rise in I moves the effective frontier and reassigns the marginal task to capital.

Cost constrained

$\tilde{I} < I$, hence $I^* = \tilde{I}$. A rise in I expands feasibility but does not change actual task assignment.

Technological possibility is not the same as effective automation.

Displacement effect

Consider a static automation-only change with fixed K :

$$dl > 0, \quad dN = 0, \quad I^* = I < \tilde{I}.$$

- ① Newly automatable marginal tasks move from labor to capital.
- ② Labor is compressed into a smaller set of tasks.
- ③ Diminishing returns within the remaining labor tasks reduce relative labor demand.

Proposition 2 gives

$$\frac{d \ln(W/R)}{dl} = -\frac{\Lambda_I}{\hat{\sigma} + \varepsilon_L} < 0.$$

Thus effective automation lowers W/R , the labor share, and employment under the paper's labor-supply specification.

Productivity effect and the wage

Automation also replaces expensive effective labor with cheaper capital:

$$d \ln Y|_{K,L} > 0.$$

For the technology-constrained allocation, Proposition 3 decomposes the wage:

$$d \ln W = d \ln Y|_{K,L} + (1 - s_L) \left[\frac{\Lambda_N}{\hat{\sigma} + \varepsilon_L} dN - \frac{\Lambda_I}{\hat{\sigma} + \varepsilon_L} dl \right].$$

With $dN = 0$:

$$\underbrace{d \ln W}_{\text{ambiguous}} = \underbrace{d \ln Y|_{K,L}}_{\text{productivity (+)}} - \underbrace{(1 - s_L) \frac{\Lambda_I}{\hat{\sigma} + \varepsilon_L} dl}_{\text{displacement (-)}}.$$

Output can rise while the wage falls.

When can automation raise the wage?

Under AER Assumptions 1–3, Proposition 3 establishes a finite threshold $\tilde{K}$:

$$K > \tilde{K} \implies \frac{d \ln W}{d l} > 0, \quad K < \tilde{K} \implies \frac{d \ln W}{d l} < 0.$$

- At low K , the cost-saving gap at the marginal task is small: displacement dominates.
- At high K , capital is abundant and the productivity gain can dominate.
- This concerns the *absolute wage*; it does not reverse the fall in W/R or the labor share.

Source discipline

Published AER version, Proposition 3, pp. 1501–1503.

Labor share and employment

Why can the wage rise while the labor share falls?

$$s_L = \frac{WL}{RK + WL}, \quad \frac{d \ln(W/R)}{dI} < 0.$$

- Productivity can increase the level of W .
- Displacement changes relative factor demand in favor of capital.
- Hence W and W/R need not move together.
- Under equation (4)'s labor-supply preferences, automation raises output per worker more than the wage, so employment falls.

The capital share rises as the complementary net-output factor share.

New tasks and reinstatement

A higher N introduces a new high-index task where labor has comparative advantage. Under Assumption 3,

$$R > \frac{W}{\gamma(N)},$$

so the new task is adopted with labor.

Change	Output	Wage	Employment	Labor share	W/R
Higher N	+	+	+	+	+

This is **reinstatement**: labor returns to a wider range of activities. It is not merely more productivity; it changes the task content of production in labor's favor.

Capital adjustment and endogenous technology

Long-run capital adjustment: Proposition 5

On the relevant interior balanced-growth path,

$$R = \rho + \delta + \theta g.$$

Capital adjustment passes productivity gains toward the relatively inelastic factor. Long-run automation raises the wage but still lowers employment and the labor share.

Direction of innovation: Proposition 6 and Corollary 2

A temporary automation imbalance can make further automation less attractive and new-task creation more attractive. A permanent shift toward automation instead leads to a new BGP with lower task content $n = N - I$, employment, and labor share.

Does automation necessarily reduce wages and labor share?

Experiment	Wage	Labor share
I rises but I^* does not	No marginal effect	No marginal effect
Effective automation, fixed K	Conditional on productivity versus displacement	Falls
Effective automation, K adjusts on BGP	Rises under Proposition 5 conditions	Falls
New tasks	Rises	Rises

Rigorous answer

No. Effective automation reduces the labor share under the relevant conditions, but its wage effect depends on productivity, displacement, the capital stock, and the adjustment horizon.

Where I did not believe the AI

AI initially wrote AER equation (5) as effective task cost divided by $1 - \eta$.

Why doubt it? That expression did not match the task technology or the displayed source equation.

AER says

$$p(i) = \min\{R, W/\gamma(i)\}^{1-\eta} \quad \text{for } i \leq I.$$

Correct interpretation The positive power preserves the cost ranking, so the allocation frontier is unchanged, but division would give wrong task-price levels.

Source: published AER, equation (5), p. 1496. This was a real transcription error, not a manufactured disagreement.

Hand derivation: wage sign under automation

Proposition 3

Para $dN=0$: $d \ln W = d \ln Y|_{K,L} - (1-s_L) \frac{\Delta_I}{\theta + \varepsilon_L} dI$
(aislamos automatización)

$$\downarrow d \ln Y|_{K,L} = P_I dI$$

$$d \ln W = \left[P_I - (1-s_L) \frac{\Delta_I}{\theta + \varepsilon_L} \right] dI$$

$$\frac{d \ln W}{dI} \gtrless 0 \iff P_I \gtrless (1-s_L) \frac{\Delta_I}{\theta + \varepsilon_L}$$

Por Proposition 3: $K > \tilde{K} \Rightarrow \frac{d \ln W}{dI} > 0$

$$K < \tilde{K} \Rightarrow \frac{d \ln W}{dI} < 0$$

Por Proposition 2: $\frac{d \ln (W/R)}{dI} < 0$

Real handwritten derivation: Proposition 3 wage threshold and Proposition 2 relative-factor-price result.

Lean formalization: the narrow claim

Paper source

NBER Working Paper 22252, revised June 2017, Section 2.2, p. 8, equation (6) and the threshold discussion immediately following it.

$$\frac{w}{r} = \gamma(c), \quad e = \min\{a, c\}, \quad i < e.$$

Lean proves the implication

$$i < \min\{a, c\} \implies i \leq a \wedge r < \frac{w}{\gamma(i)}.$$

Lean symbol	Economic meaning
a / automationThreshold	technological frontier
c / costThreshold	cost frontier
i	task index
w, r, γ	wage, rental rate, labor productivity

Lean proof and scope

Core proof logic (identifiers romanized for the slide)

```
have hiCost : i < costThreshold :=
  lt_of_lt_of_le hi (min_le_right _ _)
have hgammaLt : gamma i < w / r := by
  rw [hthreshold]
  exact hgamma hiCost
have hmul : gamma i * r < w :=
  (lt_div_iff0 hr).mp hgammaLt
apply (lt_div_iff0 (hgammaPos i)).mpr
simpa [mul_comm] using hmul
```

Assumptions

- γ strictly increasing and positive;
- $w > 0, r > 0$;
- $w/r = \gamma(c)$;
- $i < \min\{a, c\}$.

Verified: a task below both frontiers is feasible for capital and capital is strictly cheaper.

Not verified: a full proposition, equilibrium, wage effects, shares, dynamics, or balanced growth.

Actual result

Focused build passed; `paper_contribution.py check AR18RaceManMachine -fast` passed. Semantic closeout remains pending in the preserved audit scaffolding.

Conclusion

- ➊ Only a change in I^* , not capability I alone, reallocates tasks.
- ➋ Effective automation creates displacement and productivity together.
- ➌ The labor share falls, but the wage can rise or fall at fixed capital; $\tilde{K}$ separates the cases.
- ➍ New labor-intensive tasks reinstate labor and work in the opposite direction.
- ➎ Lean verifies one source-connected threshold implication—useful, but deliberately much narrower than the paper's economic propositions.

Answer

Automation does not necessarily reduce both wages and the labor share.